

Benchmarking Decoders for Directional Intent Control of a Mobile Gait Trainer Using a Dry-Electrode EEG-Based BCI

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Introduction

Recovering the ability to walk after a spinal cord injury or stroke requires repeated, intentional practice. Rehabilitation is most effective when the user actively drives the action of a task as naturally as possible. The feedback provided by executing that action in a robotic system helps retrain the damaged nervous system. To achieve this, a "dry" EEG headset can comfortably read the user's brain signals so that a decoder can infer their intent, enabling them to voluntarily control a mobile, gravity-supported gait trainer, capable of performing basic movements [1]. Therefore, this work benchmarks nine time-series-capable decoder models on a five-class (Idle, Forward, Backward, Left, Right) directional movement task, exploring which model architectures are most promising for clinical use.

Materials and Methods

Two healthy participants wore the DSI-24 wireless dry-electrode headset to record EEG signals. Each performed one cued direction per 15-second recording, yielding the labelled data used to train models to steer a wheeled gait system (Fig. 1, left). A class-balanced subset of this data formed the main benchmark, while data from a second participant was used for a cross-person test. Prior to decoding, all signals were band-pass filtered and segmented into two-second windows with a half-second stride (Fig. 1, right).

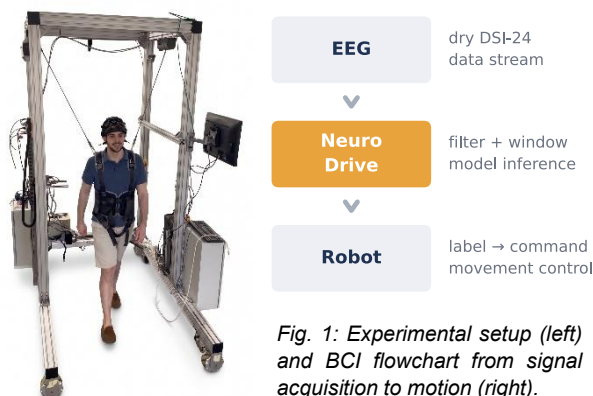


Fig. 1: Experimental setup (left) and BCI flowchart from signal acquisition to motion (right).

To evaluate performance, nine decoders were compared under shared tuning conditions. These included: a classical machine-learning pipeline (FBCSP+LDA), four models trained from scratch (compact EEGNet, TimesNet, iTransformer, and S-Mamba), three pre-trained EEG foundation models optimized for this task (Neuro-GPT, CBraMod, and LaBraM), and Chronos, which served as a non-physiological control. All were tested across three protocols: a within-subject ceiling test, a leave-recordings-out protocol, and a cross-person transfer test. Decoders were demonstrated in real time on a smaller wheeled surrogate of the gait trainer.

Results

TimesNet achieved the highest balanced accuracy, though not significantly higher than the other top models (Fig. 2), with none emerging as a clear leader [2]. EEGNet reached comparable accuracy at a fraction of the parameters and training cost. Cross-person transfer remained near chance, although live sessions on the surrogate with unseen participants performed better than expected. All decoders operated well within real-time constraints. A confidence gate raised accuracy to 94.1% on the 76% of windows it accepted. The most frequent error was forward/backward reversals in 10.3% of Forward windows.

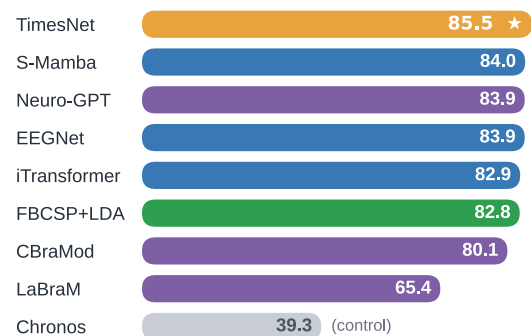


Fig. 2: Balanced accuracy (leave-recordings-out protocol).

Discussion

Several architectures reached accuracies in the low-to-mid-80% range against a 20% chance level, using augmented training data from a single participant. Since no architecture emerged as a clear winner and the best models performed within a few points of each other, size, resource requirements and inference latency can guide the choice where accuracy differences are marginal. EEG pretraining matched but did not surpass the models trained from scratch. Because cross-person transfer failed across all decoders, per-patient calibration remains necessary, although the discrepancy with live performance warrants further study. Rejecting uncertain predictions kept the device idle rather than executing an incorrect movement. For practical deployment, TimesNet with a confidence gate is recommended, as it runs in real time, with EEGNet preferable when computational resources are limited. A usable directional decoder is feasible with dry electrodes, provided each patient is calibrated individually.

References

- [1] J. Fang et al., "Development of a mobile gait trainer using pose estimation", *ICNR*, La Granja, Spain, 2025, pp. 541-545.
- [2] L. Brigato et al., "There are no champions in supervised long-term time series forecasting", *Transactions on Machine Learning Research*, 2026.